Evaluation on the improvements of transformer prediction with convolution

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***Abstract*—Proposed and developed 5 years ago, Transformer has been a prevailing machine learning method and is widely used to solve various kinds of practical problems [1]. According to relevant works, Transformer has performed well in both natural language processing and computer vision tasks, so we would like to test its effectiveness in prediction, specifically, time series prediction. Over the past two years, COVID-19 is no doubt one of the major factors that influences the changes in the stock prices, and the medical industry should be among the most significantly affected, which would provide an ideal sample for us to study transformer on time series prediction. In this paper, we not only construct a machine learning model using Transformer to try to predict the stock prices of one medical company but also add a convolution layer to try to optimize the predictions. The comparison of the outcome from the two models suggests that the convolution layer could improve the performance of the naive transformer in several ways.**

***Index Terms*—transformer, convolution, time series prediction, stock prices**

I. Introduction

The COVID-19, founded in Hubei Province in December 2019, spreaded rapidly to China, and eventually the whole world. Currently, COVID-19 has already influenced more than 100 countries as well as regions, and the number of confirmed cases has reached 560 million. During the early stage of the outbreak, many medical companies made lots of profit from selling medical supplies such as masks, thermometers ,and respirators. Nowadays, since nucleic acid testing is gradually becoming part of our routine, medical companies who provide these tests become more profitable [2] and researches on their performance are more valuable. In this paper, we first introduce related works on this topic and then present our data source as well as the structure of our models. In the third section, we show the metrics we choose to evaluate our models and discuss the performance of our models in detail. Finally, we draw a conclusion based on the results of our experiments and demonstrate several directions for future research.

Besides, rational investors have begun to re-investigate the medical industry [3]. Apart from that, balancing current growth, investment for future growth, and the delivery of shareholder return remains a major challenge for medical companies [2]. All these situations add up to the difficulty to predict the stock prices for medical companies and reduce the reliability of traditional analysis methods. So exploring new ways of performing this task is not only necessary but also practical. Since this task is closely related to data analysis, we turn to machine learning methods for a possible solution.

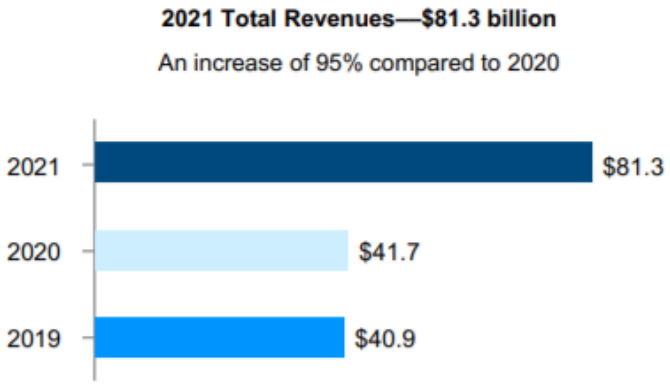

Fig. 1. The surging profit of Pfizer adapted from 2021 Annual Report [2].

II. Literature Review

A. Stock Price Prediction

Predicting the stock prices has always been a fascinating and controversial topic in the fiance area. Stock prices can be affected by many factors, such as investors' confidence, the future of the company, the change of national policies, domestic and foreign economic environments, and international situations, etc. [4], [5]. Nowadays the advancement of technology has granted the public access to a larger quantity of information, which means that stock analysis has become more and more difficult as a considerable amount of data has to be processed in a relatively short time [6]. Even though many traditional analysts cast doubt on the application of machine learning methods in this area, daily stock prices serve as a convenient data source for testing machine learning techniques, and people are hoping that making progress in big data, especially in the deep learning field, could assist them in analyzing stock information [7]. So far, there are already several studies and papers on this topic, each of which comes up with a model using somewhat different methods. Sreelekshmy Selvin et al. researched on stock price prediction using LSTM, RNN, and CNN-sliding window model, where they state that the CNN model is capable of capturing the trend of the data and has the best performance among the three

models [8]. Another research by Wenjie Lu et al. comes up with a relatively convoluted network, which consists of CNN, BiLSTM, and AM parts. Despite its complicated structure, the model performs surprisingly well at over 98 percent accuracy on the testing data set [9].

B. CNN

Since its invention in 1988, while retaining the two basic concepts of local connectedness and analog circuit dynamics, the CNN paradigm has evolved to cover a very broad class of problems and frameworks [10]. For instance, the classical type of CNN includes multi-layer perceptron (MLP) or the fully connected (FC) network, which consists of several linear layers and nonlinear activation layers [11], [12]. Particularly, CNN is a classical method in vision classification and analysis area, for it is capable of extracting meaningful features from processing shift-invariant data such as images [13], [14].

C. Transformer

Transformer is a new type of neural network that mainly make use of the self-attention mechanism [15], [16] to extract intrinsic features. First proposed by Ashish Vaswani et al., it is an incredibly powerful tool for natural language processing tasks [1]. In fact, transformer brings significant improvement in natural language processing (NLP) tasks [17], [18], [19]. So far transformer has shown great potential for extensive use in a variety of areas. The success of Transformer-based models motivates us to apply transformer to time series prediction and see if it could have a comparable performance. After building and testing the transformer model, we also tried to use convoluted layers to reduce the impact of noise brought by the original data set to achieve better performance and make a comparison. Overall, this research aims to provide a way of stock price prediction in post-COVID-19 era via transformer and presents possible methods to optimize the model for better performance.

D. Vision Transformer

A newly developed method, also known as Vision Transformer (ViT), is recently proposed by Dosovitskiy et al. [20]. and has achieved state-of-the-art performance on multiple image recognition benchmarks. The survey done by Kai Han and others suggests that transformer is a hot topic in the field of computer vision due to its competitive performance and tremendous potential compared with CNNs [17]. In fact, CNNs are considered the fundamental component in vision applications [21], [22], while transformer has become a potential alternative to CNN. Additional research by Ido Galil et al. reinforces that, among the tested methods and models, no other DNN had ever demonstrated an uncertainty estimation performance comparable—in any metric tested or setting, in-distribution or class-out-of-distribution—to the ViT architecture, which in some way indicates the supremacy of transformer methods over others, especially in uncertainty estimation [23]. Moreover, this method could also be applied to solve other related problems like object detection [24], [25], semantic segmentation [26], image processing [27], and video understanding [28].

III. Method and Model

A. Data Selection

In this paper, we chose stock price data from June 29, 2020, to June 27, 2022, a total of 503 trading days, as the research objects for analysis. Specifically, because of the lack of data on stock prices for every minute, we use the adjusted close price of the stock every day as the data set. Machine learning method like transformer has a requirement on the size of the training set and the data volume of Weekly or monthly data would be too small to subtract a large enough training set. Therefore, we choose daily data as the input and reference for the prediction result. Additionally, since we are focusing on testing and optimizing one prediction model, we select and study the stock price of Abbott Laboratories, a medical company that involves business on rapid over-the-counter COVID-19 self-testing. Abbott Laboratories was founded in 1888 and is based in North Chicago, Illinois. Together with its subsidiaries, discovers, develops, and manufactures, the company sells health care products worldwide and operates in four segments: Established Pharmaceutical Products, Diagnostic Products, Nutritional Products, and Medical Devices.

B. Transformer

Our model uses the standard architecture of Transformer, and the following is the graphic illustration.

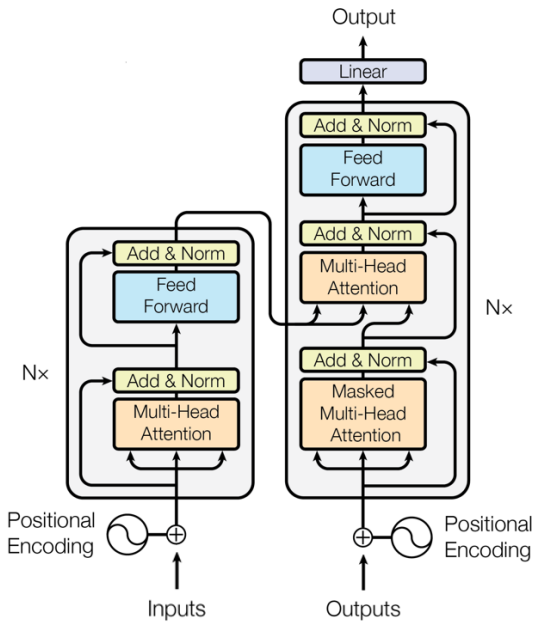

Fig. 2. The Transformer model architecture adapted from [1].

In the experiment, we first train this model on the adjusted close price of ABT. However, as we will further discuss in detail later, no matter how we modify the hyperparameters in this model, the model cannot provide a satisfying prediction, which is disappointing.

C. Convolutional Transformer

Since stock prices contain lots of noise, we hope to reduce the impact that they may have during training, and CNN is well known for its noise reduction, so we decide to add convoluted layers ahead of the positional encoding layer. The architecture of the optimized model is as the following.

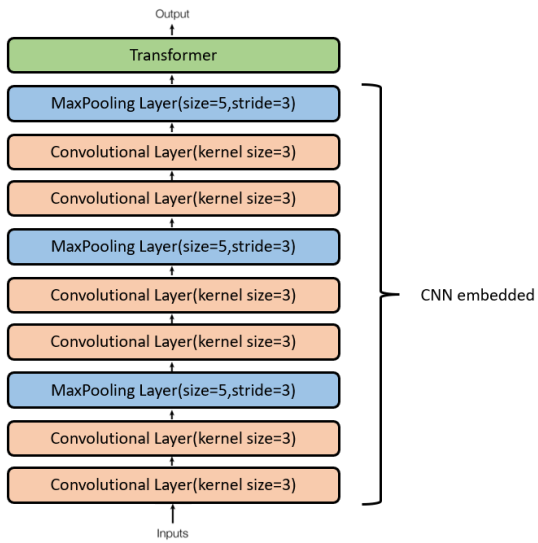

Fig. 3. The optimized Transformer model architecture.

IV. Evaluation

A. Metrics

To evaluate the prediction effect of the models, the mean absolute error (MAE), root mean square error (RMSE), root mean square error (MSE), and accuracy (ACC) are used as the evaluation metrics of the models. The MAE calculation formula is as follows:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i| \tag{1}$$

where $\hat{y}_i$ is the predicted value and y_i is the real value. The smaller the MAE is, the more accurate the prediction is.

The MSE calculation formula is as follows:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2 \tag{2}$$

where $\hat{y}_i$ is the predicted value and y_i is the real value. The smaller the MSE is, the more accurate the prediction is.

The RMSE calculation formula is as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{3}$$

where $\hat{y}_i$ is the predicted value and y_i is the real value. The smaller the RMSE is, the more accurate the prediction is.

The ACC calculation formula is as follows:

$$ACC = \frac{1}{n}\sum_{i=1}^{n}(1 - \frac{|\hat{y}_i - y_i|}{y_i}) * 100 \tag{4}$$

where $\hat{y}_i$ is the predicted value and y_i is the real value. The higher the ACC is, the more accurate the prediction is.

Following are the results of the four metrics we use to evaluate the performance of the two models.

TABLE I
Results on metrics

Model name	*MSE*	*RMSE*	*MAE*	*ACC*
Transformer	326.13	18.06	14.53	87.15
Convolutional Transformer	121.84	11.04	8.29	92.72

Our result suggests on all tested metrics, the model with convolutional layers performs better.

B. Visualization

According to the work by Dominik Sacha et al., VIS4ML is a very important part when it comes to the evaluation of training results [29]. Indeed, it is hard to say whether the performance of one model is better than another only by comparing metrics because the metrics do not tell if our model catches some important feature or pattern in prediction. So we use VIS4ML workflow to come up with a comprehensive comparison between the two models. Following is the comparison of the stock price prediction from the two models with regard to the ground truth.

The graph indicates that the optimized model is better at catching the vibration within the stock price, and thus becomes more accurate than the original transformer model, which agrees with the observation from the metrics as well.

Conclusion

In this paper, we mainly discuss the performance of transformer on the adjusted close price of Abbott Laboratories data set. Our results show that Transformer is a possible solution for time series prediction in the finance area and adding convolutional layers to reduce noise could be beneficial to the overall performance of the model. There are several directions that we would like to investigate in future research. Firstly, we will apply the model to companies from other industries to test the model's generality. Secondly, we may also expand the time range to obtain a larger sample set. Lastly, the result from the model could still be improved by doing modifications similar to adding convolutional layers.

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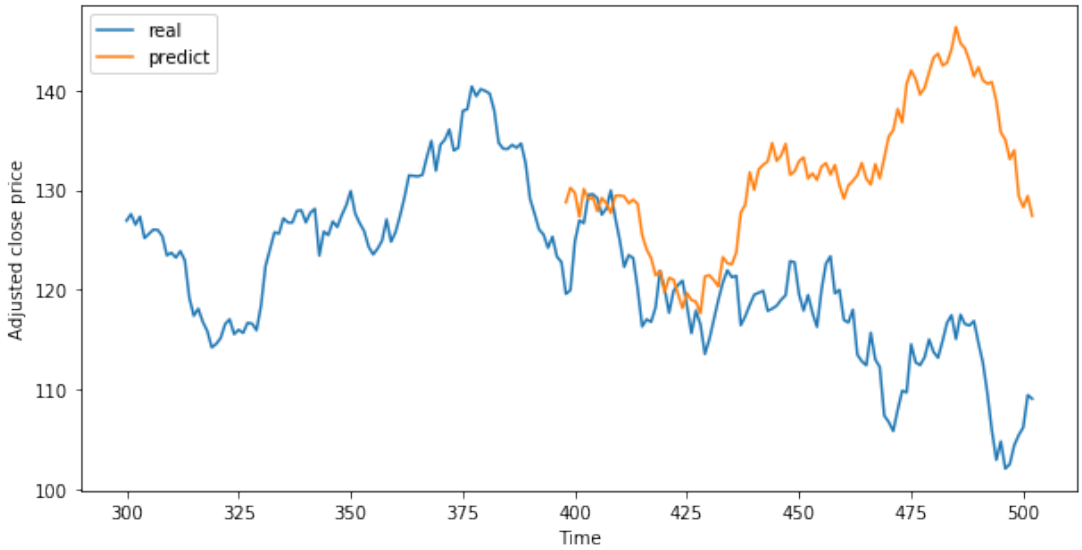

Fig. 4. Traditional transformer prediction.

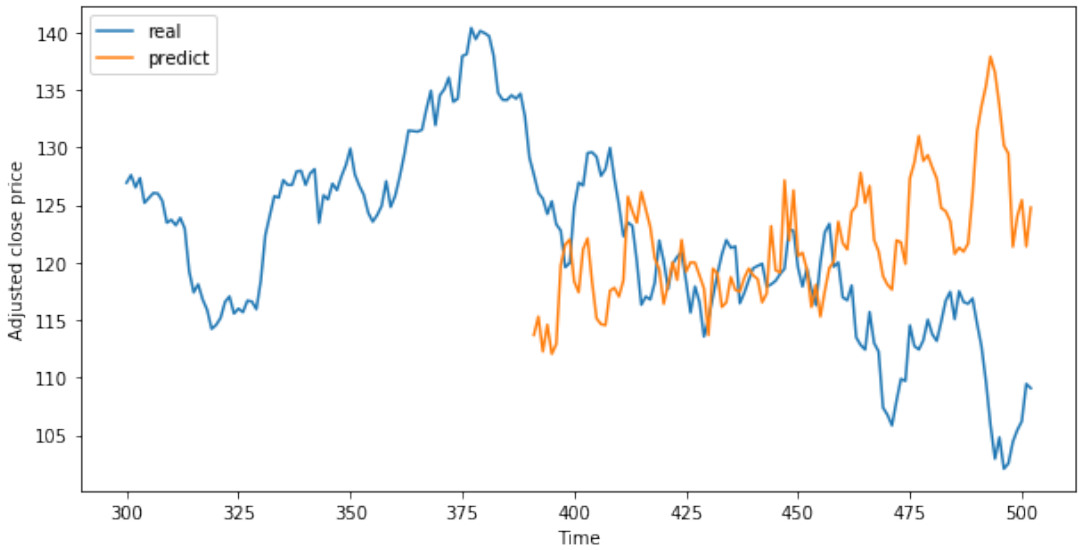

Fig. 5. Prediction from convolutional transformer with the same dataset.

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